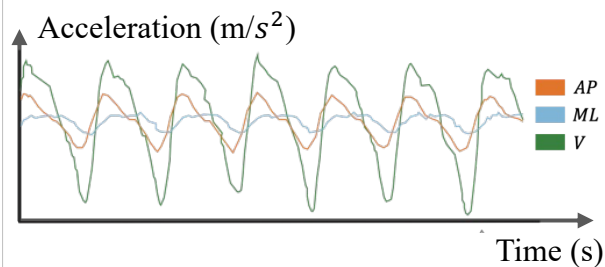


A.**Step 1**

Extract trunk gait features
from 30-m walk



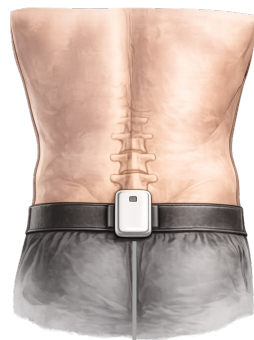
Tri-axial
trunk acceleration



Purpose:
Quantify trunk motion
during walking in PD

Step 2

Reduce features to
biomechanical domains



Domain-specific PCA

Rhythmicity
-Recurrence

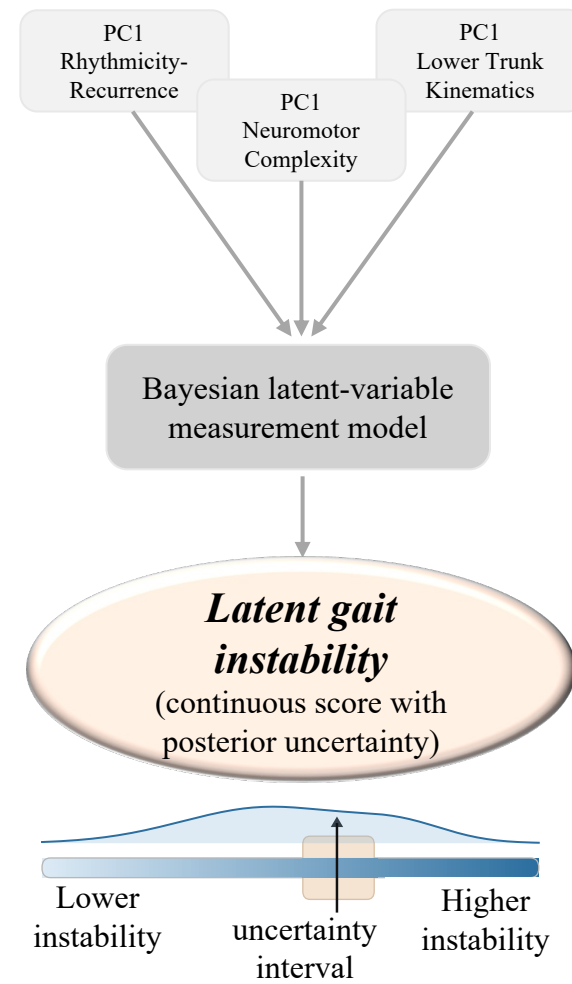
Neuromotor
Complexity

Lower Trunk
Kinematics

Purpose:
Summarize related features
into domain indicators

C.**Step 3**

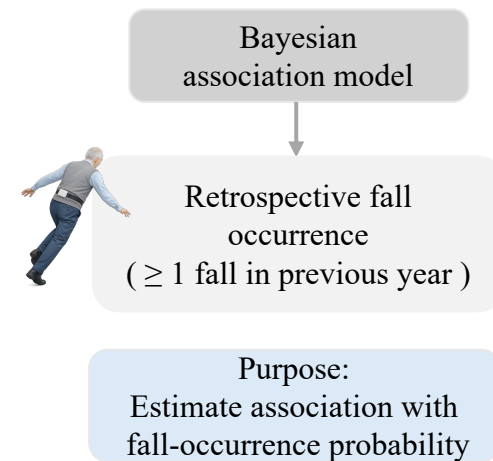
Estimate latent gait instability



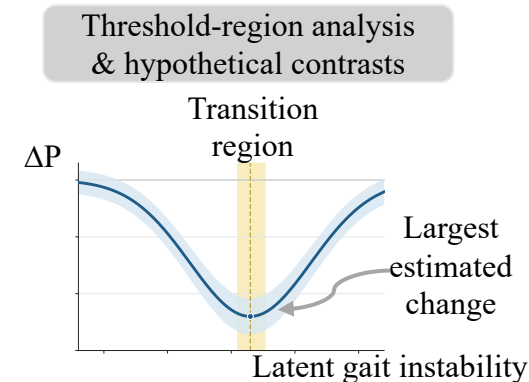
Purpose:
Turn three domain indicators into
an interpretable instability construct

D.**Step 4**

Relate instability to fall occurrence

**E.****Step 5**

Map probability changes



Purpose:
Locate where fitted probability
changes most rapidly